

ELITE MATHEMATICS

Pearson Edexcel International A Level

# Pure 2

## Adaptive Classified

Chapter 11 · Applications of Differentiation

### Questions

Student book · questions with writing space

- 68 pages
- Past papers up to January 2026

## PURE 2 | TOPIC 11

# Applications of Differentiation

Your classified route

**20**

QUESTIONS

**175**

MARKS

**4**

FAMILIES

**5**

VARIANTS

DR ESLAM AHMED

**CLASSIFIED BY IDEA**

**Move from recognition to method, then use the working space to complete the solution.**

## TOPIC 11 | FAMILY 01

# Gradient/Rate

**WHAT THIS FAMILY ASKS**

Recognise the gradient/rate demand and complete the required reasoning in a clear, exam-ready order.

**VARIANT ROUTE****V01** core family move**2 classified questions**

19 total marks

10. The curve  $C$  has equation

$$y = ax^3 - 3x^2 + 3x + b$$

where  $a$  and  $b$  are constants.

Given that

- the point  $(2, 5)$  lies on  $C$
- the gradient of the curve at  $(2, 5)$  is 7

(a) find the value of  $a$  and the value of  $b$ .

(4)

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10. The curve  $C$  has equation

$$y = ax^3 - 3x^2 + 3x + b$$

where  $a$  and  $b$  are constants.

Given that

- the point  $(2, 5)$  lies on  $C$
- the gradient of the curve at  $(2, 5)$  is 7

(b) Prove that  $C$  has no turning points.

(3)

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9.

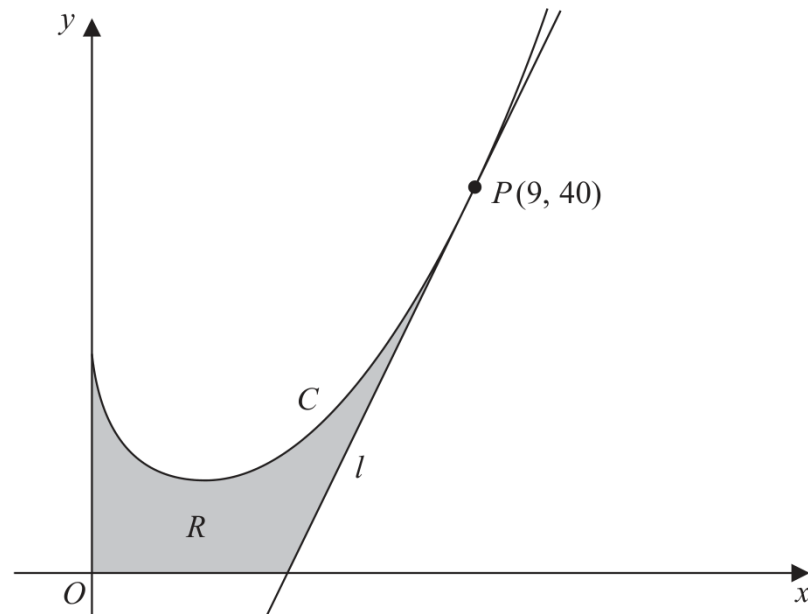


Figure 3

In this question you must show all stages of your working.

Adaptive unit 1 · WMA12-01\_2023\_Oct\_Q09

Solutions relying entirely on calculator technology are not acceptable.

Figure 3 shows a sketch of part of the curve  $C$  with equation

$$y = \frac{2}{3}x^2 - 9\sqrt{x} + 13 \qquad x \geqslant 0$$

(a) Find, using calculus, the range of values of  $x$  for which  $y$  is increasing.

(4)

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9.

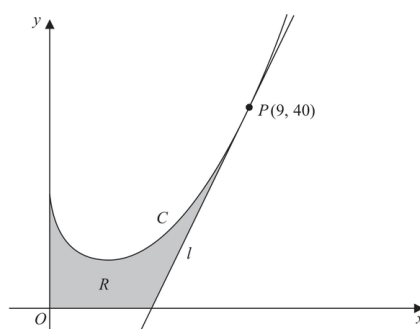


Figure 3

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

Figure 3 shows a sketch of part of the curve  $C$  with equation

$$y = \frac{2}{3}x^2 - 9\sqrt{x} + 13 \quad x \geq 0$$

The point  $P$  lies on  $C$  and has coordinates  $(9, 40)$ .

The line  $l$  is the tangent to  $C$  at the point  $P$ .

The finite region  $R$ , shown shaded in Figure 3, is bounded by the curve  $C$ , the line  $l$ , the  $x$ -axis and the  $y$ -axis.

(b) Find, using calculus, the exact area of  $R$ .

(8)

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## TOPIC 11 | FAMILY 02

# Optimisation

**WHAT THIS FAMILY ASKS**

Recognise the optimisation demand and complete the required reasoning in a clear, exam-ready order.

**VARIANT ROUTE**

**V01** Use constraints to remove extra variables first.

**V02** State the maximum/minimum value in context, not just  $x$ .

**4 classified questions**

40 total marks

9.

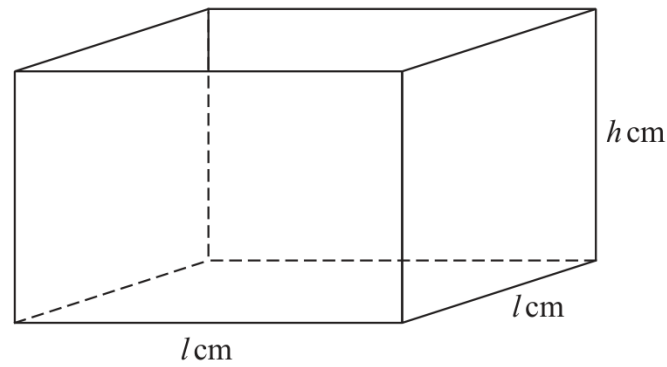
**Figure 3**

Figure 3 shows a sketch of a square based, open top box.

The height of the box is  $h$  cm, and the base edges each have length  $l$  cm.

Given that the volume of the box is  $250\,000\text{ cm}^3$

(a) show that the external surface area,  $S\text{ cm}^2$ , of the box is given by

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(3)

9.

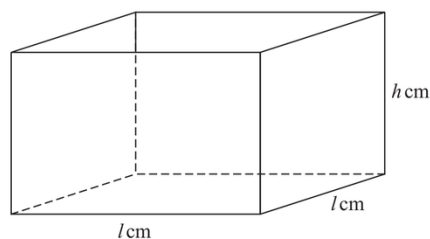


Figure 3

Figure 3 shows a sketch of a square based, open top box.

The height of the box is  $h$  cm, and the base edges each have length  $l$  cm.

Given that the volume of the box is  $250\,000\text{ cm}^3$

- (b) Use algebraic differentiation to show that  $S$  has a stationary point when  $h = 250^k$  where  $k$  is a rational constant to be found.

(5)

9.

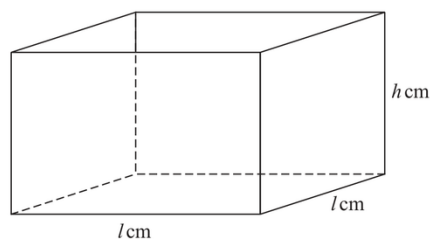


Figure 3

Figure 3 shows a sketch of a square based, open top box.

The height of the box is  $h$  cm, and the base edges each have length  $l$  cm.

Given that the volume of the box is  $250\,000\text{ cm}^3$

- (c) Justify by further differentiation that this value of  $h$  gives the minimum external surface area of the box.

(2)

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3.

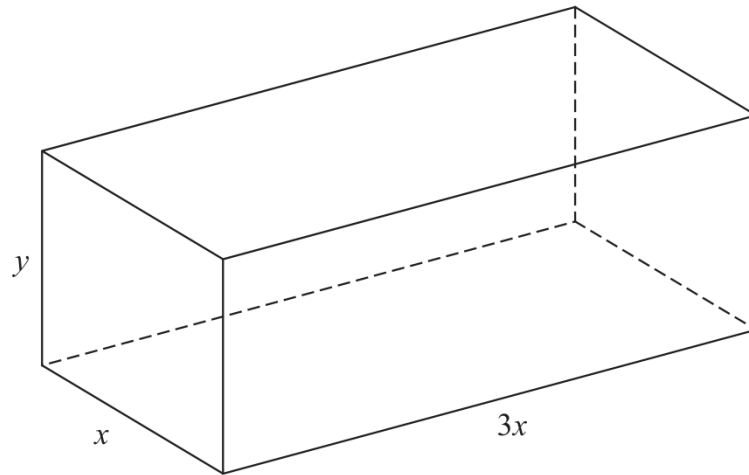
**Figure 1**

Figure 1 shows an open-topped container used for holding water.

The container is in the shape of a cuboid and is made of sheet metal.

The base of the container is a rectangle  $3x$  metres by  $x$  metres.

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The height of the container is  $y$  metres as shown in Figure 1.

Given that the capacity of the container is  $120\text{m}^3$

(a) show that the area  $A\text{ m}^2$  of the sheet metal used to make the container is given by

$$A = Px^2 + \frac{Q}{x}$$

where  $P$  and  $Q$  are positive constants to be found.

(4)

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3.

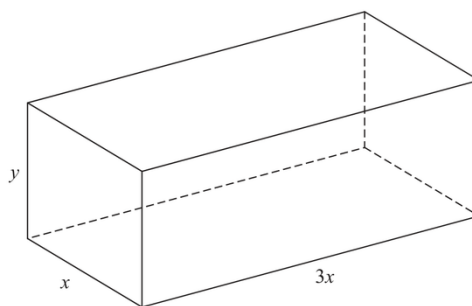
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The base of the container is a rectangle  $3x$  metres by  $x$  metres.

The height of the container is  $y$  metres as shown in Figure 1.

Given that the capacity of the container is  $120\text{ m}^3$

- (b) Use calculus to find the value of  $x$  for which  $A$  has a stationary value, giving your answer to 3 significant figures.

**(4)**

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3.

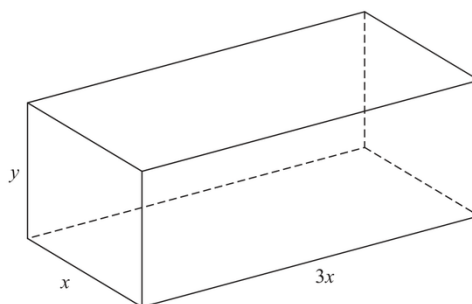


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The container is in the shape of a cuboid and is made of sheet metal.

The base of the container is a rectangle  $3x$  metres by  $x$  metres.

The height of the container is  $y$  metres as shown in Figure 1.

Given that the capacity of the container is  $120\text{ m}^3$

- (c) Find  $\frac{dA}{dx^2}$  and hence show that the value of  $x$  found in part (b) gives the minimum value of  $A$ .

(2)

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2.

In this question you must show all stages of your working.

Solutions based entirely on calculator technology are not acceptable.

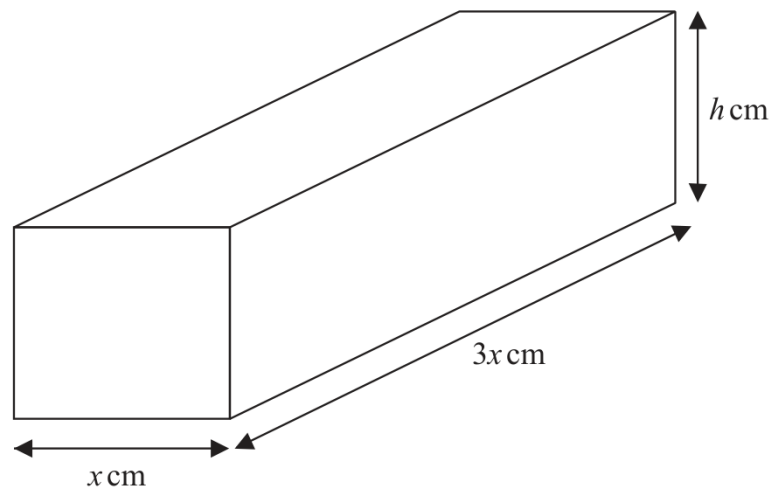


Figure 2

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A brick is in the shape of a cuboid with width  $x$  cm, length  $3x$  cm and height  $h$  cm, as shown in Figure 2.

The volume of the brick is  $972 \text{ cm}^3$

(a) Show that the surface area of the brick,  $S \text{ cm}^2$ , is given by

$$S = 6x^2 + \frac{2592}{x}$$

(3)

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2. In this question you must show all stages of your working.  
Solutions based entirely on calculator technology are not acceptable.

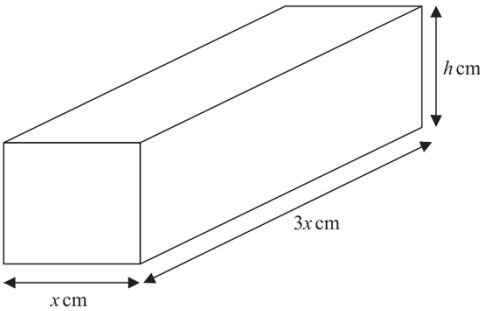


Figure 2

A brick is in the shape of a cuboid with width  $x$  cm, length  $3x$  cm and height  $h$  cm, as shown in Figure 2.  
The volume of the brick is  $972\text{ cm}^3$

(b) Find  $\frac{dS}{dx}$

(1)

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2.

In this question you must show all stages of your working.

Solutions based entirely on calculator technology are not acceptable.

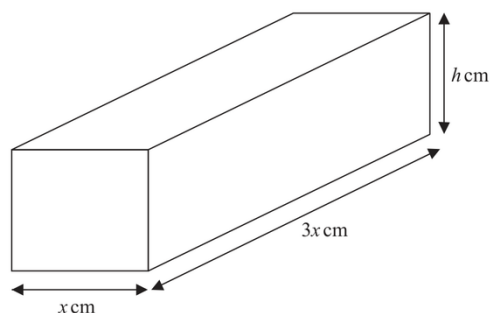


Figure 2

A brick is in the shape of a cuboid with width  $x$  cm, length  $3x$  cm and height  $h$  cm, as shown in Figure 2.

The volume of the brick is  $972 \text{ cm}^3$

(c) Hence find the value of  $x$  for which  $S$  is stationary.

(2)

Adaptive unit 3 · WMA12-01\_2023\_Jan\_Q02

2.

In this question you must show all stages of your working.

Solutions based entirely on calculator technology are not acceptable.

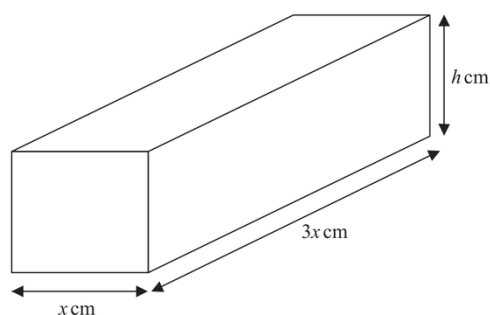


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A brick is in the shape of a cuboid with width  $x$  cm, length  $3x$  cm and height  $h$  cm, as shown in Figure 2.

The volume of the brick is  $972 \text{ cm}^3$

- (d) Find  $\frac{d^2S}{dx^2}$  and hence show that the value of  $x$  found in part (c) gives the minimum value of  $S$ .

(2)

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2. In this question you must show all stages of your working.  
Solutions based entirely on calculator technology are not acceptable.

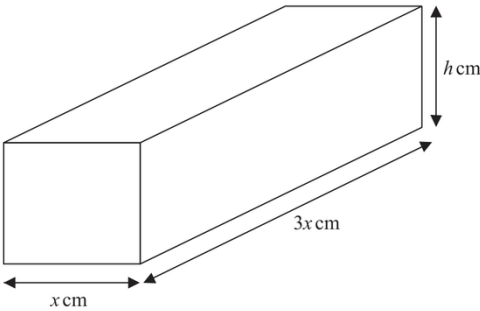


Figure 2

A brick is in the shape of a cuboid with width  $x$  cm, length  $3x$  cm and height  $h$  cm, as shown in Figure 2.  
The volume of the brick is  $972\text{ cm}^3$

(e) Hence find the minimum surface area of the brick.

(1)

9:

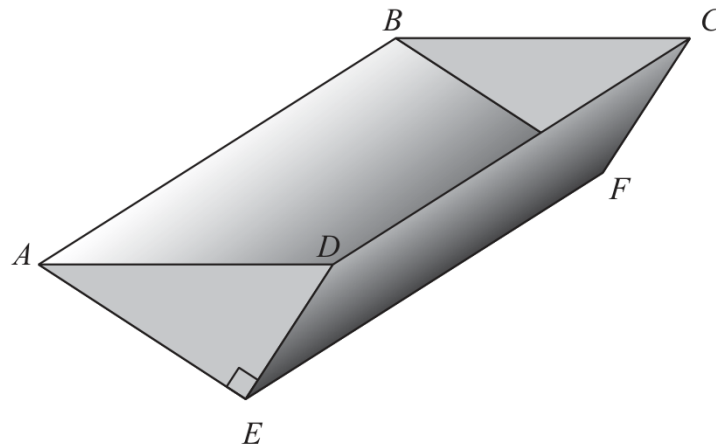
**Figure 2**

Figure 2 shows a sketch of an open container.

The sides  $CDEF$  and  $ABFE$  are rectangles.

The ends  $ADE$  and  $BCF$  are congruent (identical) right-angled triangles.

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The container is made from metal of negligible thickness.

Given that

- $AE = BF = 3x$  metres
- $DE = CF = 2x$  metres
- $AB = DC = EF = L$  metres

and the capacity of the container is  $12 \text{ m}^3$

(a) show that the area of metal used to make the container,  $S \text{ m}^2$ , is given by

$$S = Px^2 + \frac{Q}{x}$$

where  $P$  and  $Q$  are positive integers to be found.

(4)

Adaptive unit 1 · WMA12-01\_2025\_MayJune\_Q09

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9:

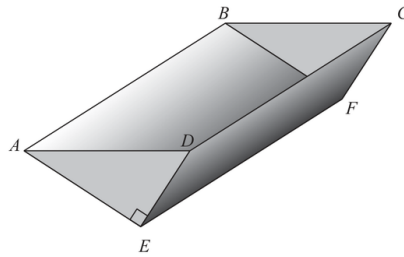


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Given that

- $AE = BF = 3x$  metres
- $DE = CF = 2x$  metres
- $AB = DC = EF = L$  metres

and the capacity of the container is  $12 \text{ m}^3$

Given that  $x$  can vary,

- (b) use algebraic calculus to find the minimum value of  $S$ , giving your answer to one decimal place.

(5)

9:

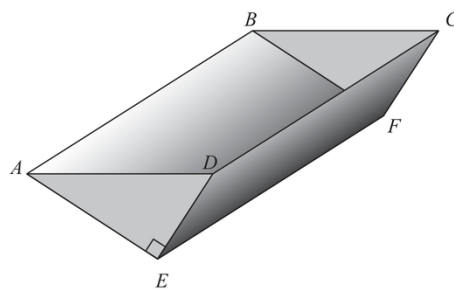


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The container is made from metal of negligible thickness.

Given that

- $AE = BF = 3x$  metres
- $DE = CF = 2x$  metres
- $AB = DC = EF = L$  metres

and the capacity of the container is  $12 \text{ m}^3$

(c) Justify that the value of  $x$  found in part (b) is a minimum.

(2)

## TOPIC 11 | FAMILY 03

# Curve Behaviour

**WHAT THIS FAMILY ASKS**

Recognise the curve behaviour demand and complete the required reasoning in a clear, exam-ready order.

**VARIANT ROUTE**

**V01** core family move

**3 classified questions**

25 total marks

- (4)

1. A curve  $C$  has equation  $y = 2x^2(x - 5)$

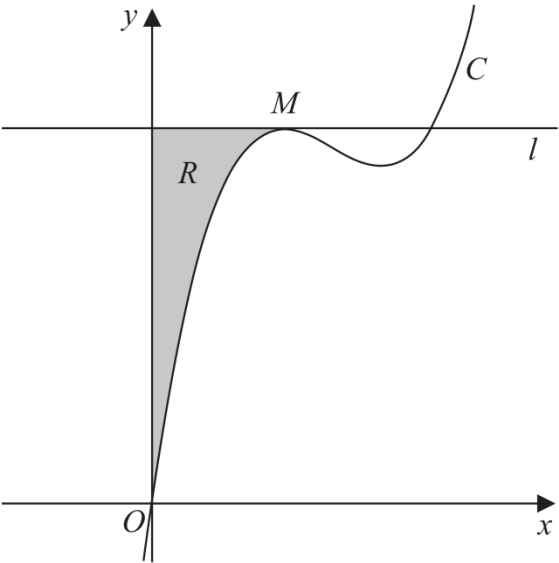
(b) Hence find the values of  $x$  for which  $y$  is increasing.

(2)

Adaptive unit 2 · WMA12-01\_2019\_Oct\_Q01

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8. In this question you must show all stages of your working.  
Solutions relying entirely on calculator technology are not acceptable.



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**Figure 2**

Figure 2 shows a sketch of part of the curve  $C$  with equation

$$y = \frac{4}{3}x^3 - 11x^2 + kx \quad \text{where } k \text{ is a constant}$$

The point  $M$  is the maximum turning point of  $C$  and is shown in Figure 2.

Given that the  $x$  coordinate of  $M$  is 2

(a) show that  $k = 28$

**(3)**

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8. In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

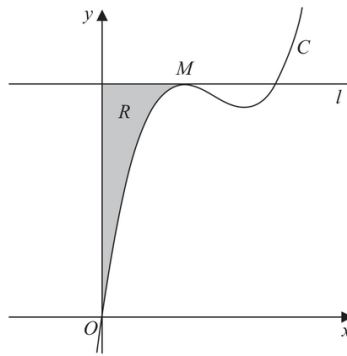


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$$y = \frac{4}{3}x^3 - 11x^2 + kx \quad \text{where } k \text{ is a constant}$$

The point  $M$  is the maximum turning point of  $C$  and is shown in Figure 2.

Given that the  $x$  coordinate of  $M$  is 2

(b) Determine the range of values of  $x$  for which  $y$  is increasing.

(2)

Adaptive unit 2 · WMA12-01\_2021\_Oct\_Q08

8. In this question you must show all stages of your working.  
Solutions relying entirely on calculator technology are not acceptable.

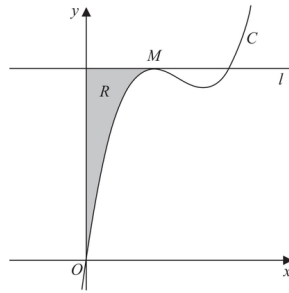


Figure 2

Figure 2 shows a sketch of part of the curve  $C$  with equation

$$y = \frac{4}{3}x^3 - 11x^2 + kx \quad \text{where } k \text{ is a constant}$$

The point  $M$  is the maximum turning point of  $C$  and is shown in Figure 2.

Given that the  $x$  coordinate of  $M$  is 2

The line  $l$  passes through  $M$  and is parallel to the  $x$ -axis.

The region  $R$ , shown shaded in Figure 2, is bounded by the curve  $C$ , the line  $l$  and the  $y$ -axis.

- (c) Find, by algebraic integration, the exact area of  $R$ .

(5)

Adaptive unit 3 · WMA12-01\_2021\_Oct\_Q08

6.

$$f(x) = 4x^3 + px^2 + 8x + q$$

where  $p$  and  $q$  are constants.

Given that

- $(2x + 3)$  is a factor of  $f(x)$
- $f(x)$  has a remainder of  $-5$  when divided by  $(x + 2)$

(a) (i) show that  $p = 10$

(ii) find the value of  $q$ .

(5)

Adaptive unit 1 · WMA12-01\_2024\_MayJuneR\_Q06

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6.

$f(x) = 4x^3 + px^2 + 8x + q$

where  $p$  and  $q$  are constants.

Given that

- $(2x + 3)$  is a factor of  $f(x)$
  - $f(x)$  has a remainder of  $-5$  when divided by  $(x + 2)$

(b) Hence find the range of values of  $x$  for which  $f(x)$  is decreasing.

*(Solutions based entirely on calculator technology are not acceptable.)*

(4)

Adaptive unit 2 · WMA12-01\_2024\_MayJuneR\_Q06

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## TOPIC 11 | FAMILY 04

# Stationary Points

**WHAT THIS FAMILY ASKS**

Recognise the stationary points demand and complete the required reasoning in a clear, exam-ready order.

**VARIANT ROUTE****V01** core family move**11 classified questions**

91 total marks

10. A curve  $C$  has equation

$$y = 4x^3 - 9x + \frac{k}{x} \quad x > 0$$

where  $k$  is a constant.

The point  $P$  with  $x$  coordinate  $\frac{1}{2}$  lies on  $C$ .

Given that  $P$  is a stationary point of  $C$ ,

(a) show that  $k = -\frac{3}{2}$

(4)

Adaptive unit 1 · WMA12-01\_2020\_Jan\_Q10

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10. A curve  $C$  has equation

$$y = 4x^3 - 9x + \frac{k}{x} \quad x > 0$$

where  $k$  is a constant.

The point  $P$  with  $x$  coordinate  $\frac{1}{2}$  lies on  $C$ .

Given that  $P$  is a stationary point of  $C$ ,

(b) Determine the nature of the stationary point at  $P$ , justifying your answer.

(2)

Adaptive unit 2 · WMA12-01\_2020\_Jan\_Q10

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10. A curve  $C$  has equation

$$y = 4x^3 - 9x + \frac{k}{x} \quad x > 0$$

where  $k$  is a constant.

The point  $P$  with  $x$  coordinate  $\frac{1}{2}$  lies on  $C$ .

Given that  $P$  is a stationary point of  $C$ ,

The curve  $C$  has a second stationary point.

(c) Using algebra, find the  $x$  coordinate of this second stationary point.

(4)

Adaptive unit 3 · WMA12-01\_2020\_Jan\_Q10

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2. A curve has equation

$$y = x^3 - x^2 - 16x + 2$$

(a) Using calculus, find the  $x$  coordinates of the stationary points of the curve.

(4)

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2. A curve has equation

$$y = x^3 - x^2 - 16x + 2$$

(b) Justify, by further calculus, the nature of all of the stationary points of the curve. **(3)**

Adaptive unit 2 · WMA12-01\_2021\_Jan\_Q02

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2. In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

The curve  $C$  has equation

$$y = 27x^{\frac{1}{2}} - x^{\frac{3}{2}} - 20 \qquad x > 0$$

(a) Find  $\frac{dy}{dx}$ , giving each term in simplest form.

(2)

Adaptive unit 1 · WMA12-01\_2022\_Jan\_Q02

Handwriting practice area with horizontal lines and a diagonal watermark reading "DR ESAM AHMED".

2. In this question you must show all stages of your working.  
Solutions relying entirely on calculator technology are not acceptable.

The curve  $C$  has equation

$$y = 27x^{\frac{1}{2}} - x^{\frac{3}{2}} - 20 \quad x > 0$$

(b) Hence find the coordinates of the stationary point of  $C$ .

(4)

Adaptive unit 2 · WMA12-01\_2022\_Jan\_Q02

DR ESAM AHMED

**In this question you must show all stages of your working.**

The curve  $C$  has equation

$$y = 27x^{\frac{1}{2}} - x^{\frac{3}{2}} - 20 \quad x > 0$$

(2)

8. In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

A curve has equation

$$y = 256x^4 - 304x - 35 + \frac{27}{x^2} \quad x \neq 0$$

(a) Find  $\frac{dy}{dx}$

(3)

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8. In this question you must show all stages of your working.  
Solutions relying entirely on calculator technology are not acceptable.

A curve has equation

$$y = 256x^4 - 304x - 35 + \frac{27}{x^2} \quad x \neq 0$$

(b) Hence find the coordinates of the stationary points of the curve.

(5)

Adaptive unit 2 · WMA12-01\_2022\_MayJune\_Q08

Handwritten solution area with horizontal lines and a diagonal watermark reading "DR ESLAM AHMED".

7. The curve  $C$  has equation

$$y = \frac{12x^3(x-7) + 14x(13x-15)}{21\sqrt{x}} \quad x > 0$$

(a) Write the equation of  $C$  in the form

$$y = ax^{\frac{7}{2}} + bx^{\frac{5}{2}} + cx^{\frac{3}{2}} + dx^{\frac{1}{2}}$$

where  $a$ ,  $b$ ,  $c$  and  $d$  are fully simplified constants.

(3)

Adaptive unit 1 · WMA12-01\_2022\_Oct\_Q07

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7. The curve  $C$  has equation

$$y = \frac{12x^3(x-7) + 14x(13x-15)}{21\sqrt{x}} \quad x > 0$$

The curve  $C$  has three turning points.

Using calculus,

(b) show that the  $x$  coordinates of the three turning points satisfy the equation

$$2x^3 - 10x^2 + 13x - 5 = 0$$

(3)

Adaptive unit 2 · WMA12-01\_2022\_Oct\_Q07

DR ESAM AHMED

$$y = \frac{12x^3(x-7) + 14x(13x-15)}{21\sqrt{x}} \quad x > 0$$

(3)

7. The height of a river above a fixed point on the riverbed was monitored over a 7-day period.

The height of the river,  $H$  metres,  $t$  days after monitoring began, was given by

$$H = \frac{\sqrt{t}}{20}(20 + 6t - t^2) + 17 \quad 0 \leq t \leq 7$$

Given that  $H$  has a stationary value at  $t = \alpha$

- (a) use calculus to show that  $\alpha$  satisfies the equation

$$5\alpha^2 - 18\alpha - 20 = 0$$

(5)

Adaptive unit 1 · WMA12-01\_2023\_MayJune\_Q07

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The height of the river,  $H$  metres,  $t$  days after monitoring began, was given by

$$H = \frac{\sqrt{t}}{20}(20 + 6t - t^2) + 17 \qquad 0 \leq t \leq 7$$

Given that  $H$  has a stationary value at  $t = \alpha$

(b) Hence find the value of  $\alpha$ , giving your answer to 3 decimal places.

(1)

Adaptive unit 2 · WMA12-01\_2023\_MayJune\_Q07

DR ESAM AHMED

7. The height of a river above a fixed point on the riverbed was monitored over a 7-day period.

The height of the river,  $H$  metres,  $t$  days after monitoring began, was given by

$$H = \frac{\sqrt{t}}{20}(20 + 6t - t^2) + 17 \qquad 0 \leq t \leq 7$$

Given that  $H$  has a stationary value at  $t = \alpha$

(c) Use further calculus to prove that  $H$  is a maximum at this value of  $\alpha$ .

(2)

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10.

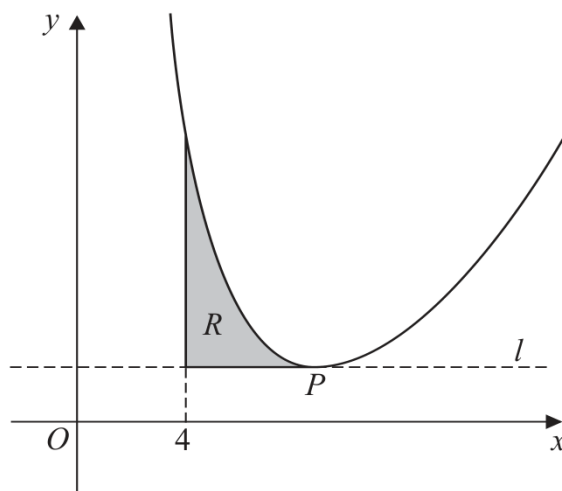
**In this question you must show detailed reasoning.****Solutions relying entirely on calculator technology are not acceptable.****Figure 2**

Figure 2 shows a sketch of the curve with equation

Adaptive unit 1 · WMA12-01\_2024\_Jan\_Q10

$$y = \frac{1}{2}x^2 + \frac{1458}{\sqrt{x^3}} - 74 \quad x > 0$$

(a) Use calculus to show that the  $x$  coordinate of  $P$  is 9

(4)

Adaptive unit 1 · WMA12-01 2024 Jan Q10

10.

In this question you must show detailed reasoning.

Solutions relying entirely on calculator technology are not acceptable.

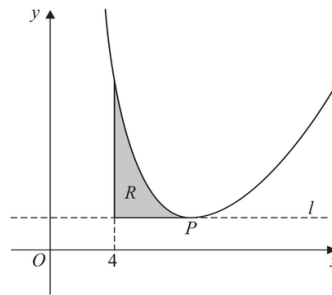


Figure 2

Figure 2 shows a sketch of the curve with equation

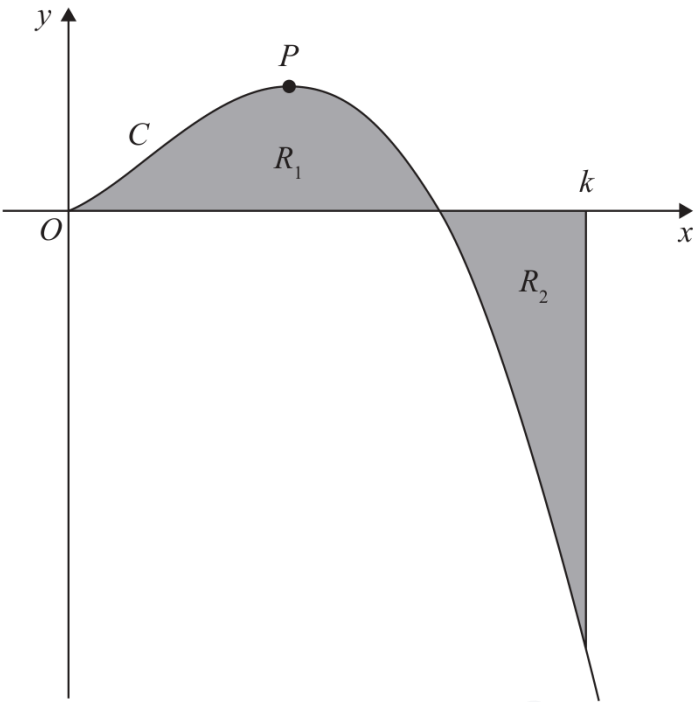
$$y = \frac{1}{2}x^2 + \frac{1458}{\sqrt{x^3}} - 74 \quad x > 0$$

The point  $P$  is the only stationary point on the curve.The line  $l$  passes through the point  $P$  and is parallel to the  $x$ -axis.The region  $R$ , shown shaded in Figure 2, is bounded by the curve, the line  $l$  and the line with equation  $x = 4$ (b) Use algebraic integration to find the exact area of  $R$ .

(5)

Adaptive unit 2 · WMA12-01\_2024\_Jan\_Q10

9.



Adaptive unit 1 · WMA12-01\_2024\_MayJune\_Q09

Figure 1

Figure 1 is a sketch of the curve  $C$  with equation

$$y = 2x^{\frac{3}{2}}(4 - x) \qquad x \geqslant 0$$

The point  $P$  is the stationary point of  $C$ .

(a) Find, using calculus, the  $x$  coordinate of  $P$ .

(4)

Adaptive unit 1 · WMA12-01\_2024\_MayJune\_Q09

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9.

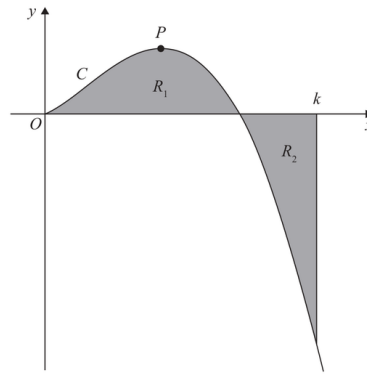


Figure 1

Figure 1 is a sketch of the curve  $C$  with equation

$$y = 2x^{\frac{3}{2}}(4-x) \quad x \geq 0$$

The point  $P$  is the stationary point of  $C$ .

The region  $R_1$ , shown shaded in Figure 1, is bounded by  $C$  and the  $x$ -axis.

The region  $R_2$ , also shown shaded in Figure 1, is bounded by  $C$ , the  $x$ -axis and the line with equation  $x = k$ , where  $k$  is a constant.

Given that the area of  $R_1$  is equal to the area of  $R_2$

(b) find, using calculus, the exact value of  $k$ .

(4)

Adaptive unit 2 · WMA12-01\_2024\_MayJune\_Q09

10.

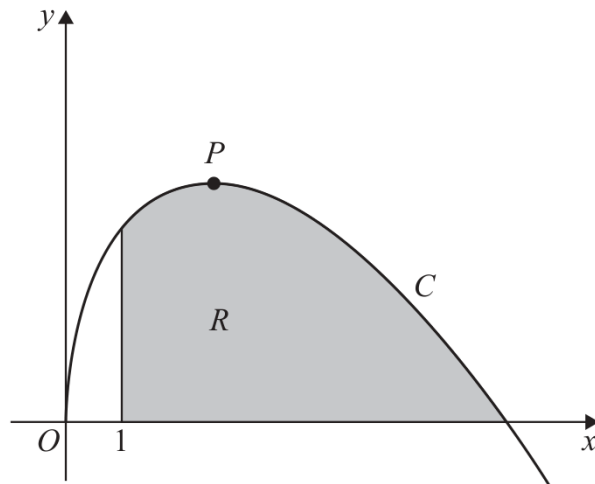


Figure 1

**In this question you must show all stages of your working.  
Solutions relying entirely on calculator technology are not acceptable.**

Figure 1 shows a sketch of part of the curve  $C$  with equation

Adaptive unit 1 · WMA12-01\_2024\_MayJuneR\_Q10

$$y = \frac{9x - x^2}{2\sqrt{x}} \qquad x > 0$$

The point  $P$  is the stationary point on  $C$ .

(a) Find, using calculus, the  $x$  coordinate of  $P$ .

(4)

Adaptive unit 1 · WMA12-01\_2024\_MayJuneR\_Q10

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10.

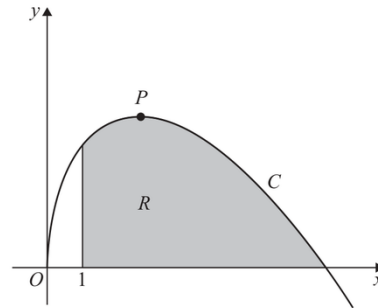


Figure 1

In this question you must show all stages of your working.  
Solutions relying entirely on calculator technology are not acceptable.

Figure 1 shows a sketch of part of the curve  $C$  with equation

$$y = \frac{9x - x^2}{2\sqrt{x}} \quad x > 0$$

The point  $P$  is the stationary point on  $C$ .

The finite region  $R$ , shown shaded in Figure 1, is bounded by the curve  $C$ , the  $x$ -axis and the line with equation  $x = 1$

(b) Using calculus, calculate the exact area of  $R$ .

(5)

Adaptive unit 2 · WMA12-01\_2024\_MayJuneR\_Q10

4. **In this question you must show all stages of your working.**  
**Solutions relying entirely on calculator technology are not acceptable.**

The curve  $C$  has equation

$$y = 4x^{\frac{1}{2}} + 9x^{-\frac{1}{2}} + 3 \qquad x > 0$$

(a) Find  $\frac{dy}{dx}$  giving each term in simplest form.

(2)

Adaptive unit 1 · WMA12-01\_2024\_Oct\_Q04

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4. In this question you must show all stages of your working.  
Solutions relying entirely on calculator technology are not acceptable.  
The curve  $C$  has equation

$$y = 4x^{\frac{1}{2}} + 9x^{-\frac{1}{2}} + 3 \quad x > 0$$

(b) Hence find the  $x$  coordinate of the stationary point of  $C$ .

(2)

Adaptive unit 2 · WMA12-01\_2024\_Oct\_Q04

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4. In this question you must show all stages of your working.  
Solutions relying entirely on calculator technology are not acceptable.  
The curve  $C$  has equation

$$y = 4x^{\frac{1}{2}} + 9x^{-\frac{1}{2}} + 3 \quad x > 0$$

- (c) (i) Find  $\frac{d^2y}{dx^2}$  giving each term in simplest form.  
(ii) Hence determine the nature of the stationary point of  $C$ , giving a reason for your answer.

(2)

Adaptive unit 3 · WMA12-01\_2024\_Oct\_Q04

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4. In this question you must show all stages of your working.  
Solutions relying entirely on calculator technology are not acceptable.  
The curve  $C$  has equation

$$y = 4x^{\frac{1}{2}} + 9x^{-\frac{1}{2}} + 3 \quad x > 0$$

(d) State the range of values of  $x$  for which  $y$  is decreasing.

(1)

Adaptive unit 4 · WMA12-01\_2024\_Oct\_Q04

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10.

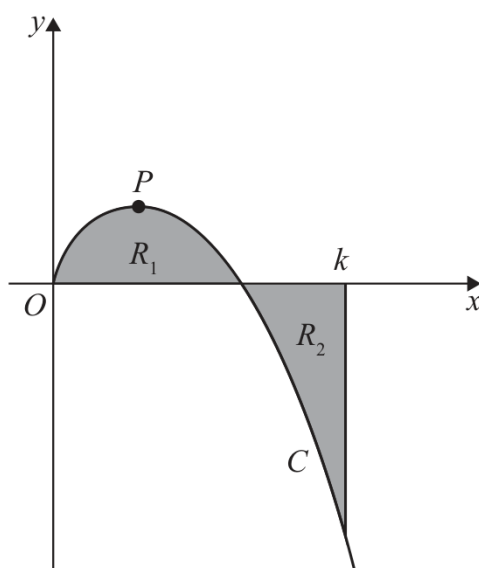


Figure 2

In this question you must show all stages of your working.

Adaptive unit 1 · WMA12-01\_2026\_Jan\_Q10

Solutions relying entirely on calculator technology are not acceptable.

Figure 2 shows a sketch of the curve  $C$  with equation

$$y = \frac{\sqrt{x}(100 - x^2)}{40} \quad x \geq 0$$

The point  $P$  is a stationary point on  $C$ .

(a) Use algebraic differentiation to find the exact  $x$  coordinate of  $P$ .

(4)

Adaptive unit 1 · WMA12-01\_2026\_Jan\_Q10

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10.

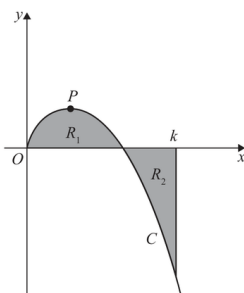


Figure 2

In this question you must show all stages of your working.  
Solutions relying entirely on calculator technology are not acceptable.

Figure 2 shows a sketch of the curve  $C$  with equation

$$y = \frac{\sqrt{x}(100 - x^2)}{40} \quad x \geq 0$$

The point  $P$  is a stationary point on  $C$ .

The region  $R_1$ , shown shaded in Figure 2, is bounded by  $C$  and the  $x$ -axis.

The region  $R_2$ , also shown shaded in Figure 2, is bounded by  $C$ , the  $x$ -axis and the line with equation  $x = k$ , where  $k$  is a constant.

Given that the area of  $R_1$  is equal to the area of  $R_2$

(b) use algebraic integration to find the exact value of  $k$ .

(4)

Adaptive unit 2 · WMA12-01\_2026\_Jan\_Q10